

Flow C - Scheduled AWS IAM Hygiene Monitoring with TheHive 5 Alert Creation

Scheduled credential-report analysis, Slack digesting, finding creation, DataTable tracking, and TheHive alert generation for IAM hygiene issues

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Flow C AWS IAM Hygiene TheHive5 Alerts

Document Type

Workflow project document | Flow C

Prototype Focus

AWS IAM automation | n8n | TheHive 5 | Slack

Project Character

Portfolio prototype with implementation, testing, and validation evidence

Design Goal

Showcase an end-to-end SOC automation workflow with clear business and analyst value

Why this document matters

- This write-up is designed to work both as a project portfolio document and as an implementation-facing prototype note.
- It explains the problem, the workflow design, the operational value to a SOC, and the validation evidence that proves the workflow worked in the lab.
- The screenshots included throughout the document are the actual workflow, node, Slack, DataTable, AWS, and TheHive validation artifacts collected during the build process.

Key outcomes

- Runs on a schedule to generate and evaluate the AWS IAM credential report.
- Detects hygiene gaps such as missing MFA, stale keys, and unused keys.
- Publishes a Slack digest, writes issue records to DataTable, and creates TheHive 5 alerts for analyst follow-up.

1. Project Summary

Flow C is the proactive hygiene and exposure-monitoring workflow of the prototype. Unlike Flow A and Flow B, which respond to event-driven findings, Flow C continuously checks the identity estate for conditions that are risky even when no active GuardDuty finding exists yet.

The workflow uses the AWS IAM credential report to detect issues such as users without MFA, stale access keys, and unused keys. Those issues are then turned into an analyst-friendly digest, a structured DataTable record, and individual TheHive alerts. This shifts the prototype from reactive handling into recurring preventive hygiene monitoring.

2. Problem Statement

Event-driven detections are important, but many IAM risks exist before a cloud detection service raises a specific incident. Weak hygiene - such as missing MFA or long-lived unused keys - quietly increases attack surface over time. Without scheduled checks, these conditions remain hidden until they contribute to an actual incident.

Why the problem matters in a SOC

- It supports preventive monitoring, not only reactive incident handling.
- It gives the SOC and cloud security team a recurring view of IAM exposure conditions.
- It creates a documented backlog of identity hygiene issues that can be tracked like other security work.

3. Objective

- Generate the AWS IAM credential report on a schedule.
- Wait for report readiness and decode the returned base64 content into CSV.
- Evaluate users against hygiene thresholds such as MFA, stale keys, and unused keys.
- Summarize the run in Slack and create finding records for the identified issues.
- Open or reuse TheHive alerts for actionable hygiene findings.

4. Solution Overview

Flow C uses a scheduled trigger to initialize configuration and thresholds, requests the credential report, waits for report completion, and then retrieves the generated report content. The workflow decodes the content, parses the CSV, and calculates hygiene findings against configurable logic such as `require_mfa`, `stale_key_days`, and `unused_key_days`.

The output is split into two levels. First, a Slack digest summarizes the overall run: account, region, users checked, and issue counts. Second, each individual issue is normalized into a finding object that can be written to DataTable and checked against existing TheHive alerts before a new alert is opened.

This design makes Flow C useful for both portfolio demonstration and practical preventive security monitoring because it produces strategic digesting and case-ready operational findings at the same time.

5. Tools and Technologies Used

Tool / Technology	Why it was used in this workflow
AWS IAM GenerateCredentialReport / GetCredentialReport	Provide the credential-report data set required for account-wide identity hygiene analysis.
n8n Wait node	Introduces the necessary report-readiness pause before the report is retrieved.
JavaScript code nodes	Decode the report, parse the CSV, calculate issue types, and build normalized findings.
Slack	Receives a digest summarizing the run and the top hygiene issues.
n8n DataTable	Persists the per-finding records so issues can be tracked across runs.
TheHive 5 native nodes	Search for existing hygiene alerts and create new ones only when required.

6. How the System Works

Stage	What happens
Scheduled start	Initializes run configuration and hygiene thresholds.
Report generation	Starts the AWS IAM credential report generation task.
Wait + retrieval	Waits for report readiness, then retrieves the base64-encoded report.
Decode + evaluate	Parses the report and identifies hygiene findings.
Digesting	Builds the summary message for Slack.
Finding split	Explodes the findings into individual records for tracking and alert creation.
TheHive alert stage	Searches for an existing alert and creates one if no match is found.

Plain-English view

- In plain English, Flow C is a recurring IAM health check for the environment.
- It asks AWS for the credential report, looks for risky identity hygiene conditions, and then turns the results into analyst-readable outputs instead of leaving them buried inside CSV data.

7. Workflow and Architecture Evidence

The final version of Flow C combines scheduled hygiene analysis with TheHive alerting. The full workflow view shows the original digesting path as well as the later alert-creation branch that operates on each finding.

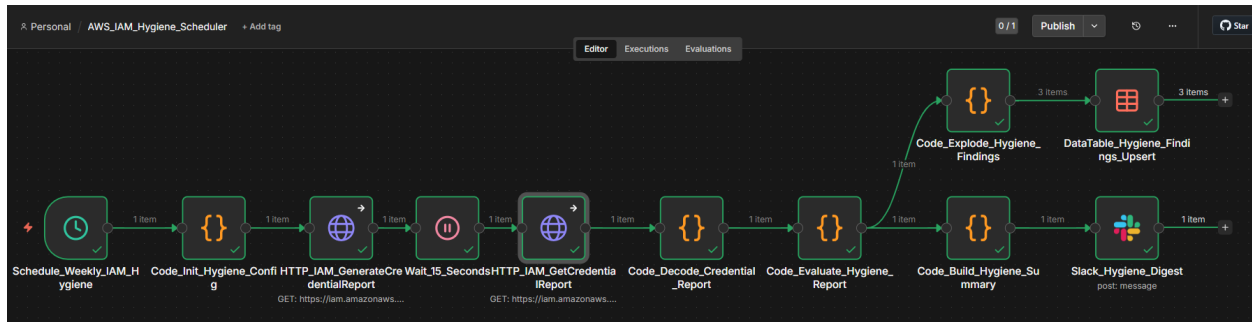


Figure C1. Full Flow C workflow with digesting and TheHive alert creation.

The screenshots below isolate the two most important structural parts of the workflow: the scheduled credential-report path and the per-finding TheHive branch.



Figure C2. Credential-report generation, wait, retrieval, and evaluation path.

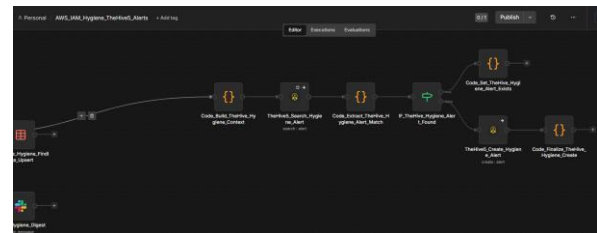


Figure C3. Per-finding TheHive search and alert creation branch.

The implementation required several iterations to solve report readiness, base64/CSV parsing, and duplicate alert logic. The screenshots below capture the working logic used in the final design.

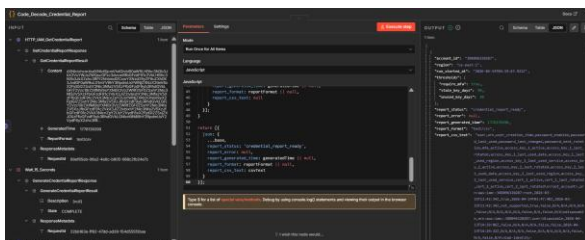


Figure C4. Report decode and extraction logic.

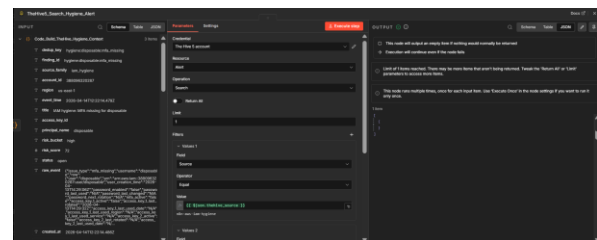


Figure C5. Native TheHive search logic for hygiene findings.

8. Key Features and USP

- Scheduled proactive IAM hygiene monitoring instead of event-only response.
- Support for configurable thresholds such as missing MFA and key-age conditions.
- Digest-style Slack output for quick operational awareness.
- Per-finding DataTable state tracking and dedup-friendly issue records.
- TheHive alert search-before-create logic to reduce duplicate alerts.

9. SOC Value and Expected Impact

Flow C adds preventive cloud-security value to the prototype. It helps the SOC or cloud security team identify IAM hygiene drift before those issues become full incidents. Operationally, this supports better

exposure management, creates a visible remediation backlog, and ensures recurring identity hygiene work is documented inside the same response tooling used for incident-driven activity.

Operational area	Expected effect
Preventive coverage	Improved by identifying risky IAM hygiene conditions on a schedule.
Analyst workload	Reduced because the report is interpreted automatically instead of reviewed manually.
Remediation tracking	Improved through per-finding DataTable rows and TheHive alerts.
Consistency	Improved because the same evaluation logic runs on every scheduled cycle.
Exposure awareness	Improved because account hygiene state is surfaced regularly in Slack.

10. Validation and Testing Evidence

A key validation milestone was solving the credential-report retrieval sequence. Once the wait-and-retrieve pattern worked, the workflow was able to decode the report and evaluate real hygiene findings.

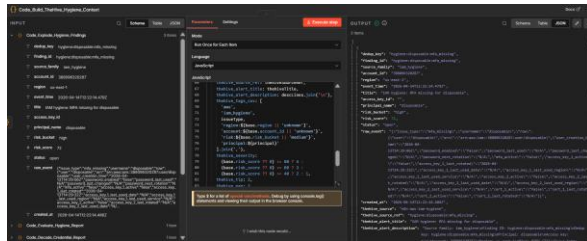


Figure C6. Report retrieval node working after the wait stage.

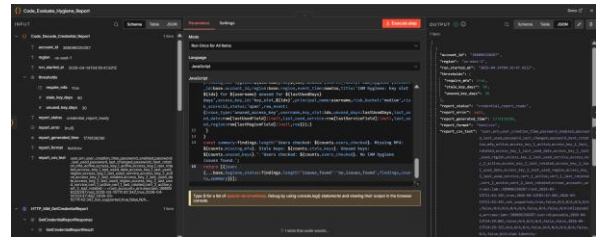


Figure C7. Evaluation logic node used to classify hygiene findings.

The Slack digest below shows the run-level output after the report was processed successfully. In the demonstrated lab run, the workflow surfaced three missing-MFA issues.

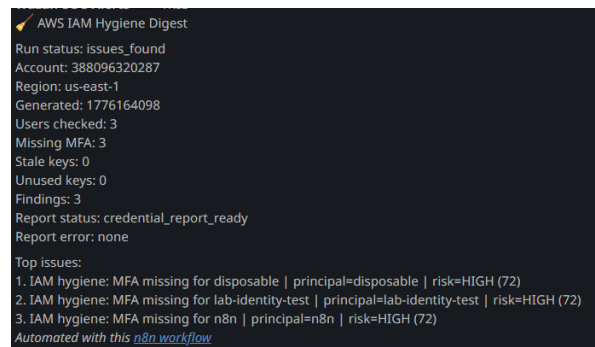


Figure C8. Final Slack digest summarizing the IAM hygiene run.

The workflow also wrote issue rows to DataTable so the hygiene findings became durable records rather than one-time chat notifications.

T dedup_key	T finding_id	T source_family	T account_id	T region	event_time	T title
Null	Null	Null	Null	Null	Null	Null
guardduty-gd-finding-001	gd-finding-001	guardduty	123456789012	us-east-1	2026-03-17T06:00:00.000-04:00	GuardDuty test finding
guardduty-gd-finding-001	gd-finding-001	guardduty	123456789012	us-east-1	2026-03-17T06:00:00.000-04:00	GuardDuty test finding
guardduty-gd-finding-001	gd-finding-001	guardduty	123456789012	us-east-1	2026-03-17T06:00:00.000-04:00	GuardDuty test finding
guardduty-gd-finding-001	gd-finding-001	guardduty	123456789012	us-east-1	2026-03-17T06:00:00.000-04:00	GuardDuty test finding
guardduty-gd-finding-001	gd-finding-001	guardduty	123456789012	us-east-1	2026-03-17T06:00:00.000-04:00	GuardDuty test finding
guardduty-gd-finding-001	gd-finding-001	guardduty	123456789012	us-east-1	2026-03-17T06:00:00.000-04:00	GuardDuty test finding
hygiene:disposable:mfa_missing	hygiene:disposable:mfa_missing	iam_hygiene	388096320287	us-east-1	2026-04-14T00:20:03.117-04:00	IAM hygiene: MFA missing for disposab
hygiene:lab-identity-test:mfa_missing	hygiene:lab-identity-test:mfa_missing	iam_hygiene	388096320287	us-east-1	2026-04-14T00:20:03.117-04:00	IAM hygiene: MFA missing for lab-ident
hygiene:n8n:mfa_missing	hygiene:n8n:mfa_missing	iam_hygiene	388096320287	us-east-1	2026-04-14T00:20:03.117-04:00	IAM hygiene: MFA missing for n8n

Figure C9. DataTable rows created for the IAM hygiene findings.

Finally, the workflow created analyst-facing alerts in TheHive 5 for the identified hygiene issues. This turns a scheduled hygiene problem into an actionable security work item.

Status	Severity	Title	# Case	Type	Source	Reference	Details	Assignee	Dates
New	risk:high	IAM hygiene: MFA missing for disposable	1	external	n8n-aws-iam-hygiene	Observables	1	?	O. 14/04/2026 17:52
					hygiene:disposable...	TTPs	0		C. 14/04/2026 17:52
									U. 14/04/2026 17:52

Figure C10. TheHive alert list showing the created IAM hygiene alert.

11. Real-World Use Cases

- Recurring IAM hardening programs in cloud security operations.
- Security teams that want preventive visibility into MFA adoption and key hygiene.
- Portfolio demonstrations of scheduled cloud security automation beyond incident response.

12. Skills Demonstrated

- Scheduled automation design in n8n.
- AWS IAM credential-report handling, wait-based orchestration, and CSV parsing.
- Per-finding normalization and duplicate-aware alert creation.
- Preventive security monitoring design and evidence-driven documentation.

13. Prototype Limitations

- The report depends on IAM credential-report data, so the workflow is limited to what that report exposes.
- Thresholds are practical defaults and may need tuning per environment.
- The prototype creates alerts for findings but does not yet automate remediation actions for hygiene drift.

14. Future Enhancements

- Add case promotion or ticket creation for unresolved hygiene findings that persist across multiple runs.
- Correlate hygiene findings with recent activity, policy context, or business owner metadata.
- Expand the logic to support additional AWS identity misconfiguration classes beyond the credential report.

15. Conclusion

Flow C extends the project from reactive security automation into recurring preventive monitoring. It proves that scheduled AWS IAM hygiene checks can be orchestrated, explained, tracked, and escalated with the same discipline used for event-driven findings. That gives the overall prototype a more mature and complete security-operations story. *Prepared by* **abdul4rehman215**